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SUPERSEDING
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MILITARY SPECIFICATION

FUZE, BASE DETONATING, M91A2 METAL PARTS FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the assembly drawings of the metal parts of one type of fuze designated as Fuze, Base Detonating, M91A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-P-10025 - Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for
MIL-C-45662 - Calibration System Requirements

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions

FSC: 1390

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MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures for Inspection by Attributes

DRAWINGS

U.S. ARMY

8837308 - Fuze, Base Detonating, M91A2 Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Parts. - The parts shall comply with all requirements specified on Drawing (dwg.) 8837308 and with all requirements specified in applicable specifications.

3.3 In-Line Position of Firing Pin. - The peripheral edge of the firing pin point shall not wobble or rotate beyond the limits of a 26 degree maximum cone whose diameter, at a location corresponding to the greatest length of the percussion plunger assembly, is .075 inch maximum when tested as specified in 4.4.2.

3.4 Embrittlement of Springs (Spring, Slider; Spring Retaining and Spring, Safety Pin). - None of the springs used in the fabrication of the fuze shall break when extended or distorted to the point of minimum permanent distortion.

3.5 Arming and Non-Arming (Slider)

3.5.1 Non-Arming. - The slider shall not arm when spun at 2400 revolutions per minute (rpm) minimum (min.).

3.5.2 Arming. - The slider shall arm when spun at 3600 rpm maximum.

3.6 Workmanship. - All parts shall be fabricated and finished in a thorough, workmanlike manner and all manufacturing,

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processing and assembling operations shall be correctly performed. All parts shall be clean and free of burrs, sharp edges, unblended radii, surface defects, cracks, chips, slivers, cuts, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

3.7 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define the terms herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government Verification. - Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government Inspector shall verify at unscheduled intervals, all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified as "Critical" be accepted solely on the basis of the contractor's records.

4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. The first article shall consist of twenty-five (25) complete sets of parts and assemblies (i.e., 25 each of every component and every subassembly), except that quantities for the following shall be:

a. Two-hundred twenty-five (225) percussion plunger assemblies.
b. Two-hundred twenty-five (225) assemblies, each consisting of body, slider spring, retaining screw and slider which is loaded with an inert material for the charge.

The first article shall have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor

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shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, including statements of findings/certificates of conformance for materials (see 4.1.1.c.). A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be Performed. - Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and be inspected to determine compliance with any or all requirements of the applicable drawings.

4.2.3 *Rejection. - If any component or assembly fails to* comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection Provisions

4.3.1 Lot Formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

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4.3.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Housing, Plunger (see dwg. 8837317 covering a detail of dwg. 8837308).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.5 Percent		
201.	Total length	Gage	01001
202.	Base thickness	Gage	01002
203.	Outside diameter	Gage	01003
204.	Inside diameter	Gage	01004
205.	Diameter of hole	Gage	01005
206.	Evidence of poor workmanship (see 3.6)	Visual	01006

4.3.2.2 Fulcrum, Firing Pin (see dwg. 8837321 covering a detail of dwg. 8837308).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.40 Percent		
101.	Diameter, max.	Gage	02001
102.	Plating damaged exposing base metal	Visual	02002
Minor:	AQL 1.0 Percent		
201.	Length	Gage	02003
202.	Evidence of poor workmanship (see 3.6)	Visual	02004

4.3.2.3 Pin, Firing (see dwg. 8837318 covering a detail of dwg. 8837308).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 Percent		
101.	Profile of firing pin point ..	Gage	03001
102.	Height of firing pin point, max.	Gage	03002
103.	Diameter of fulcrum pin hole, min.	Gage	03003
104.	Location of fulcrum pin hole ..	Gage	03004
105.	Alignment of firing pin point with fulcrum pin hole ..	Gage	03005
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	03006

4.3.2.4 Pin, Safety (see dwg. 8837320 covering a detail of dwg. 8837308).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Large diameter, max.	Gage	04001
102.	Total length, max.	Gage	04002

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	103.	Length of small diameter shaft	Gage	04003
Minor:		AQL 0.65 Percent		
	201.	Evidence of poor workmanship (see 3.6)	Visual	04004

4.3.2.5 Body, Plunger (see dwg. 8837316 covering a detail of dwg. 8837308).

Categories		Defects	Method of Inspection	Code No.
Critical:	None defined.			
Major:		AQL 0.40 Percent		
	101.	Distance to bottom of firing pin slot, max.	Gage	05001
	102.	Location of firing pin fulcrum hole	Gage	05002
Minor:		AQL 1.5 Percent		
	201.	Total length	Gage	05003
	202.	Width of firing pin slot	Gage	05004
	203.	Chamfer missing	Visual	05005
	204.	Evidence of poor workmanship (see 3.6)	Visual	05006

4.3.2.6 Percussion Plunger Assembly (see dwg. 8837315 covering a detail of dwg. 8837308).

Categories		Defects	Method of Inspection	Code No.
Critical:	None defined.			
Major:		AQL 1.0 Percent		
	101.	Total length including firing pin in armed position	Gage	06001
	102.	Total length of assembly in unarmed position	Gage	06002

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103.	Firing pin point damaged	Gage	06003
104.	Any component missing or improperly assembled	Visual	06004
Minor:	AQL 2.5 Percent		
201.	Stake of firing pin fulcrum missing	Visual	06005
202.	Stake of body pin missing	Visual	06006
203.	Plating damaged exposing base metal	Visual	06007
204.	Coating of plunger housing incomplete	Visual	06008
205.	Evidence of poor workmanship (see 3.6)	Visual	06009

4.3.2.7 Spring, Restraining (see dwg. 8837325 covering a detail of dwg. 8837308).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 Percent		
101.	Load at test height, max.	Gage	07001
102.	Load at assembled height, min.	Gage	07002
103.	Solid height, max.	Gage	07003
104.	Plating damaged exposing base metal	Visual	07004
105.	Evidence of poor workmanship (see 3.6)	Visual	07005
Minor:	AQL 1.5 Percent		
201.	Outside diameter, max.	Gage	07006
202.	Inside diameter, min.	Gage	07007
203.	Winding not left hand	Visual	07008

4.3.2.8 Holder, Detonator (see dwg. 8837323 covering a detail of dwg. 8837308).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Categories	Defects	Method of Inspection	Code No.
Major:	AQL 1.0 Percent		
101.	Pitch diameter of thread	Gage	08001
102.	Major diameter of thread	Gage	08002
103.	Length of thread pilot, min...	Gage	08003
104.	Inside diameter of firing pin hole	Gage	08004
105.	Concentricity of firing pin hole with pitch diameter of thread	Gage	08005
106.	Diameter of detonator cavity, min.	Gage	08006
107.	Depth of detonator cavity	Gage	08007
108.	Total length	Gage	08008
109.	Thread not full	Visual	08009
Minor:	AQL 2.5 Percent		
201.	Diameter of thread pilot	Gage	08010
202.	Location of wrench holes	Gage	08011
203.	Depth of wrench hole	Gage	08012
204.	Wrench hole missing	Visual	08013
205.	Evidence of poor workmanship (see 3.6)	Visual	08014

4.3.2.9 Slider (see dwg. 8837314 covering a detail of dwg. 8837308).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 Percent		
101.	Length of boss, max.	Gage	09001
102.	Distance from shoulder to charge cavity	Gage	09002
103.	Diameter of charge cavity, min.	Gage	09003
104.	Diameter of pilot	Gage	09004
105.	Alignment of pilot with shoulder	Gage	09005
106.	Alignment of charge cavity with pilot	Gage	09006
107.	Length of shoulder, min.	Gage	09007
108.	Finish improper	Visual	09008

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Minor:		AQL 4.0 Percent		
201.	Total length, max.		Gage	09009
202.	Length of shoulder, max.		Gage	09010
203.	Diameter of shoulder		Gage	09011
204.	Diameter of boss		Gage	09012
205.	Length of boss, min.		Gage	09013
206.	Diameter of gas check cavity .		Gage	09014
207.	Depth of gas check cavity, min.		Gage	09015
208.	Distance from shoulder to gas check cavity		Gage	09016
209.	Alignment of gas check cavity with pilot and charge cavity .		Gage	09017
210.	Radii missing		Visual	09018
211.	Evidence of poor workmanship (see 3.6)		Visual	09019

4.3.2.10 Spring, Slider (see dwg. 8837322 covering a detail of dwg. 8837308).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Solid height, max.	Gage	10001
102.	Plating damaged exposing base metal	Visual	10002
103.	Evidence of poor workmanship (see 3.6)	Visual	10003
Minor:	AQL 1.5 Percent		
201.	Outside diameter, max.	Gage	10004
202.	Inside diameter, min.	Gage	10005
203.	Winding not left hand	Visual	10006

4.3.2.11 Screw, Retaining (see dwg. 8837312 covering a detail of dwg. 8837308).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 1.0 Percent		
101.	Depth of slider shoulder counterbore	Gage	11001
102.	Depth of spring cavity, min.	Gage	11002
103.	Distance from open end to end of thread, min.	Gage	11003
104.	Length of thread pilot	Gage	11004
105.	Pitch diameter of thread, max.	Gage	11005
106.	Major diameter of thread, max.	Gage	11006
107.	Thread not full	Visual	11007
Minor:	AQL 4.0 Percent		
201.	Total length	Gage	11008
202.	Pitch diameter of thread, min.	Gage	11009
203.	Major diameter of thread, min.	Gage	11010
204.	Diameter of slider shoulder counterbore	Gage	11011
205.	Diameter of spring cavity	Gage	11012
206.	Depth of spring cavity, max.	Gage	11013
207.	Diameter of thread pilot	Gage	11014
208.	Distance between wrench notches	Gage	11015
209.	Wrench notch missing	Visual	11016
210.	Finish improper	Visual	11017
211.	Length of chamfer	Gage	11018
212.	Evidence of poor workmanship (see 3.6)	Visual	11019

4.3.2.12 Head (see dwg. 8837331 covering a detail of dwg. 8837308).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 2.5 Percent		
101.	Pitch diameter of shell thread	Gage	12001
102.	Major diameter of shell thread	Gage	12002

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103.	Pitch diameter of body thread.	Gage	12003
104.	Major diameter of body thread.	Gage	12004
105.	Pitch diameter of tracer thread	Gage	12005
106.	Minor diameter of tracer thread, min.	Gage	12006
107.	Distance from start of shell thread to flange, min.	Gage	12007
108.	Shell thread undercut, max.	Gage	12008
109.	Concentricity of body thread with shell thread	Gage	12009
110.	Concentricity of flange with shell thread	Gage	12010
111.	Concentricity of percussion plunger cavity with shell thread	Gage	12011
112.	Perpendicularity of body thread with face of flange	Gage	12012
113.	Perpendicularity of body thread with plunger cavity end	Gage	12013
114.	Depth of percussion plunger cavity, min.	Gage	12014
115.	Diameter of percussion plunger cavity	Gage	12015
116.	Distance from plunger cavity end to flange, min.	Gage	12016
117.	Diameter of flange	Gage	12017
118.	Thickness of flange	Gage	12018
119.	Distance from plunger cavity end to shoulder, min.	Gage	12019
120.	Length of perfect tracer thread, min.	Gage	12020
121.	Caulking groove angle missing	Visual	12021
122.	Metal defects (crack or break)	Visual	12022
123.	Shell thread not full	Visual	12023

Minor: AQL 4.0 Percent

201.	Distance from start of shell thread to flange, max.	Gage	12024
202.	Diameter of shell thread pilot, max.	Gage	12025
203.	Body thread undercut, max.	Gage	12026
204.	Length of body thread pilot	Gage	12027
205.	Diameter of body thread pilot, max.	Gage	12028

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206.	Distance from plunger cavity end to shoulder, max.	Gage	12029
207.	Depth of plunger cavity, max..	Gage	12030
208.	Diameter of caulking groove at flange	Gage	12031
209.	Length of caulking groove	Gage	12032
210.	Concentricity of tracer thread with shell thread	Gage	12033
211.	Total length	Gage	12034
212.	Distance across wrench slots .	Gage	12035
213.	Wall thickness between tracer and plunger cavity, min.	Gage	12036
214.	Tracer thread not full	Visual	12037
215.	Plating damaged exposing base metal	Visual	12038
216.	Evidence of poor workmanship (see 3.6)	Visual	12039

4.3.2.13 Body (see dwg. 8837310 covering a detail of dwg. 8837308).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 2.5 Percent		
101.	Pitch diameter of head thread	Gage	13001
102.	Minor diameter of head thread, min.	Gage	13002
103.	Pitch diameter of detonator holder thread	Gage	13003
104.	Minor diameter of detonator holder thread, min.	Gage	13004
105.	Pitch diameter of retaining screw thread	Gage	13005
106.	Minor diameter of retaining screw thread, min.	Gage	13006
107.	Pitch diameter of booster cup thread	Gage	13007
108.	Major diameter of booster cup thread	Gage	13008

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109.	Outside diameter, max.	Gage	13009
110.	Length of perfect head thread, min.	Gage	13010
111.	Depth of head cavity	Gage	13011
112.	Depth to bottom of detonator cavity from shoulder of head cavity	Gage	13012
113.	Depth to bottom of detonator cavity from shoulder of holder cavity	Gage	13013
114.	Length of perfect detonator holder thread, min.	Gage	13014
115.	Concentricity of head thread with outside diameter including perpendicularity with bottom of head cavity	Gage	13015
116.	Concentricity of head thread with detonator cavity	Gage	13016
117.	Concentricity of detonator holder thread with outside diameter including perpendicularity with bottom of detonator holder cavity	Gage	13017
118.	Distance from bottom of head cavity to centerline of slider cavity	Gage	13018
119.	Distance from body centerline to retainer screw thread undercut, max.	Gage	13019
120.	Concentricity of booster cup thread with outside diameter .	Gage	13020
121.	Diameter of slider cavity, min.	Gage	13021
122.	Total length	Gage	13022
123.	Alignment of slider cavity with slider shoulder cavity ..	Gage	13023
124.	Finish of slider cavity improper	Visual	13024
125.	Material defective (crack or splits)	Visual	13025

Minor:	AQL 4.0 Percent		
201.	Diameter of detonator cavity .	Gage	13026
202.	Diameter of lead hole	Gage	13027

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203.	Diameter of slider shoulder cavity	Gage	13028
204.	Depth of slider shoulder cavity	Gage	13029
205.	Diameter of slider cavity, max.	Gage	13030
206.	Depth of slider cavity	Gage	13031
207.	Depth of retaining screw thread cavity, min.	Gage	13032
208.	Distance from booster end to shoulder, min.	Gage	13033
209.	Diameter of booster cup thread pilot, max.	Gage	13034
210.	Length of booster cup thread pilot, min.	Gage	13035
211.	Booster cup thread undercut, max.	Gage	13036
212.	Diameter of booster cavity ...	Gage	13037
213.	Depth of booster cavity to datum diameter	Gage	13038
214.	Plating damaged exposing base metal	Visual	13039
215.	Staking notch missing or incorrectly located	Visual	13040
216.	Evidence of poor workmanship (see 3.6)	Visual	13041

4.3.2.14 Cup, Booster (see dwg. 8837313 covering a detail of dwg. 8837308).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 Percent		
101.	Outside diameter, max.	Gage	14001
102.	Pitch diameter of thread, min.	Gage	14002
103.	Length of perfect thread, min.	Gage	14003
104.	Thickness at bottom, max.	Gage	14004
105.	Concentricity of outside diameter with threads	Gage	14005
106.	Crack or break	Visual	14006

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Minor:	AQL 2.5 Percent		
201.	Outside diameter, min.	Gage	14007
202.	Pitch diameter of thread, max.	Gage	14008
203.	Minor diameter of thread, max.	Gage	14009
204.	Thickness at bottom, min.	Gage	14010
205.	Depth to bottom	Gage	14011
206.	Finish improper	Visual	14012
207.	Evidence of poor workmanship (see 3.6)	Visual	14013

4.3.2.15 Spring, Safety Pin (see dwg. 8837326 covering a detail of dwg. 8837308).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Solid height, max.	Gage	15001
102.	Evidence of poor workmanship (see 3.6)	Visual	15002
Minor:	AQL 1.5 Percent		
201.	Outside diameter, max.	Gage	15003
202.	Inside diameter, min.	Gage	15004
203.	Winding not left hand	Visual	15005

4.3.3 Testing

4.3.3.1 Arming Test of Percussion Plunger Assembly (see dwg. 8837315) - Major Defect - Code No. 06010. - Sampling plans for this test shall be in accordance with MIL-STD-105, Code Letter L and an AQL of 0.25 percent. Testing shall be performed as specified in 4.4.2.

4.3.3.2 Non-Arming Test of Percussion Plunger Assembly (see dwg. 8837315) - Critical Defect - Code No. 06011. - This test shall be performed 100 percent on the lot and also on the samples which were subjected to the arming time test (4.3.3.1). If any plunger assembly fails to comply with the requirements specified

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on the applicable drawing, the assembly shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.1.

4.3.3.3 Arming and Non-Arming Test of Slider

4.3.3.3.1 Non-Arming Test (see 3.5.1) - Critical Defect - Code No. 09020. - The sampling plan for this test shall be in accordance with MIL-STD-105, Single Sampling, Code Letter L and an AQL of .065 percent. Bodies, sliders, slider springs and retaining screws shall be selected from each lot of parts and assembled using an inert material for the charge. The test shall be conducted as specified in 4.4.3.

4.3.3.3.2 Arming Test (see 3.5.2) - Major Defect - Code No. 09021. - Using the samples specified in 4.3.3.3.1, the items shall be tested as specified in 4.4.4. An AQL of 0.25 percent shall apply to this test.

4.3.3.4 Embrittlement of Springs - Major Defects - Safety Pin (dwg. 8837326) - Code No. 15006; Restraining (dwg. 8837325) - Code No. 07009 and Slider (dwg. 8837322) - Code No. 10007. - A sample of 50 springs shall be selected from the lot by the Government Inspector. If one or more springs in any of the samples fail to comply with 3.4 when tested as specified in 4.4.5, that lot of springs shall be rejected.

4.3.4 Inspection Equipment. - Inspection equipment required to perform the inspection prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein or approved Contractor Design (6.3.2). See 6.3 for details concerning responsibilities for inspection equipment design and approval.

4.3.4.1 Government Rights to Documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Government Use of Contractor's Inspection and Test Equipment. - The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating

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the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.3.4.3 Calibration. - The contractor shall have available a set of master gages, standards and appropriate instruments for regularly scheduled calibrations of his acceptance inspection equipment. Records of such regularly scheduled calibrations shall be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments shall be in accordance with Specification MIL-C-45662.

4.4 Test Methods and Procedures

4.4.1 Non-Arming of Percussion Plunger Assembly. - Each percussion plunger assembly in the lot shall be subjected to this test using equipment specified in 4.3.4. Each assembly shall not arm when spun at, and at not less than, the non-arming speed of rotation specified on the applicable drawing. Non-arming of the percussion plunger assembly shall be evidenced by observing that the firing pin has not been released from the safety pin locked position to the armed (ready-to-fire) position.

4.4.2 Arming of Percussion Plunger Assembly. - Each assembly shall arm when spun at, and not exceeding, the arming speed speed specified on the applicable drawing, Arming shall be evidenced by rotation of the firing pin on the fulcrum to the armed (ready-to-fire) position. The peripheral edge of the firing pin point shall not wobble or rotate beyond the limits of a 26 degree maximum cone whose diameter, at a location corresponding to the greatest length of the percussion plunger assembly, is .075 inch maximum. Check manually to see that the firing pin lock functions. Percussion plungers tested for arming shall subsequently be tested for non-arming as specified in 4.4.1. Equipment shall be as specified in 4.3.4.

4.4.3 Non-Arming Test of Slider. - The slider shall not arm when the body assembly is spun at, and at not less than, the non-arming speed of rotation specified in 3.5.1. Test equipment specified in 4.3.4 shall be used. Non-arming shall be determined by observation from the head end of the body that the inert charge in the slider is not visible through the detonator flash hole.

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4.4.4 Arming Test of Slider. - The body with inert loaded slider, slider spring and retaining screw shall be rotated at the arming speed specified in 3.5.2. Arming shall be determined by observation from the head end of the body that the inert charge of the slider is in line with the detonator flash hole. Equipment specified in 4.3.4 shall be used.

4.4.5 Spring embrittlement. - The spring shall be stretched along its longitudinal axis, in the case of a coil spring, or bent in the case of a flat spring, until the spring no longer returns to its original length or profile, it shall be considered distorted permanently. Any spring that breaks prior to this point shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C. - The metal parts for Fuze, Base Detonating, M91A2 shall be preserved and packaged in accordance with Specification MIL-P-10025.

5.2 Packing

5.2.1 Level C. - The metal parts for Fuze, Base Detonating, M91A2 shall be packed in accordance with Specification MIL-P-10025.

5.3 Marking. - Marking shall be in accordance with Specification MIL-P-10025.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Code Numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs. - Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) and contractor designs. SIE designs are designated

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by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided and used by the contractor. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE Designs. - SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-1000.

c. An envelope drawing, as defined in MIL-STD-1000, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with SIE designs, if not otherwise covered thereby (see 6.3.1.c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

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6.3.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-F, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of First Article Samples. - Instructions as to the location for the evaluation of the first article sample shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-F, Dover, New Jersey 07801.

6.5 Defect Reporting. - From each lot produced, the contractor shall submit two copies of the Metal Parts Lot Acceptance Test Data to Picatinny Arsenal, ATTN: SARPA-QA-A-F and SARPA-AD-F-E within three (3) days after completion of acceptance testing of the lot. If fuzes and/or tests, in addition to acceptance requirements, are utilized to evaluate the lot, the results of these tests will also be submitted three (3) days after the additional tests are completed. The defective fuzes shall be saved pending review of the results.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1390-A-062

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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